

NETWORK TOPOLOGIES

LEARNING OBJECTIVES

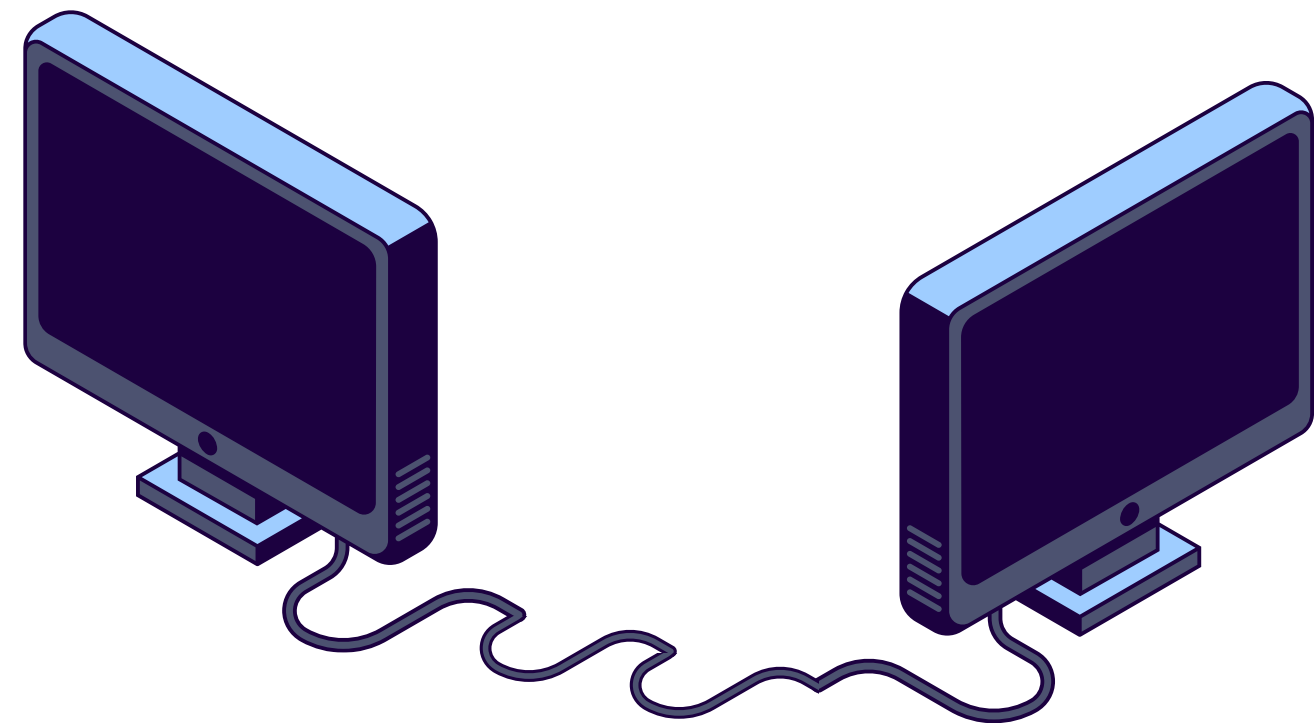
Describe common LAN topologies.

Describe the differences between the various topologies.

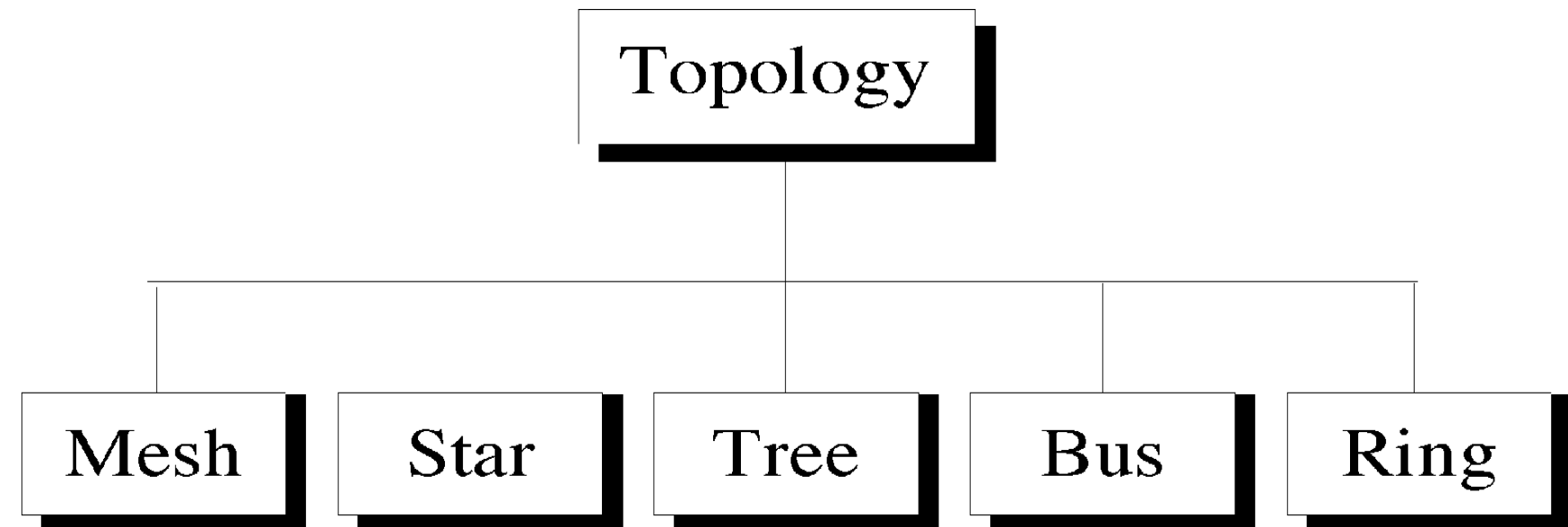
Select the most appropriate topology for a given scenario.

NETWORK TOPOLOGIES

- A network topology describes the configuration of a network (how the network components are connected together).
- The topology of a n/w is the geometric representation of the relationship of all the links & linking devices to one another.
- There are FIVE main topology's.



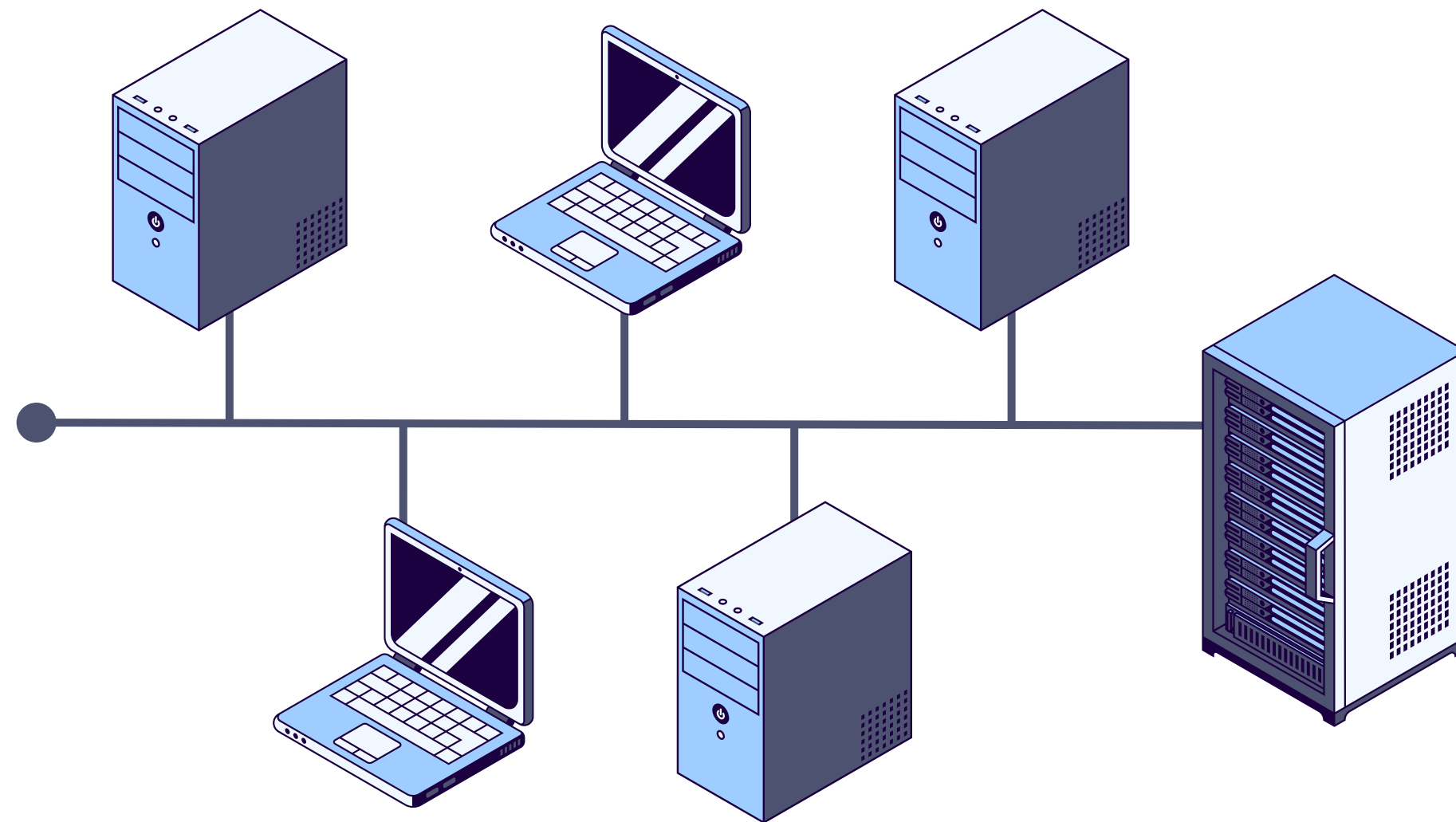
NETWORK TOPOLOGIES



Factors for selecting Topologies

- ❖ Desired Performance
- ❖ Desired Reliability
- ❖ Size(No of nodes)
- ❖ Expandability of the system
- ❖ Cost of the components
- ❖ Delays involved in routing

In a bus topology, every device is connected to a central cable (backbone). All network data is sent up and down this cable until it reaches the correct device (node).



BUS TOPOLOGY

USES FOR BUS TOPOLOGY

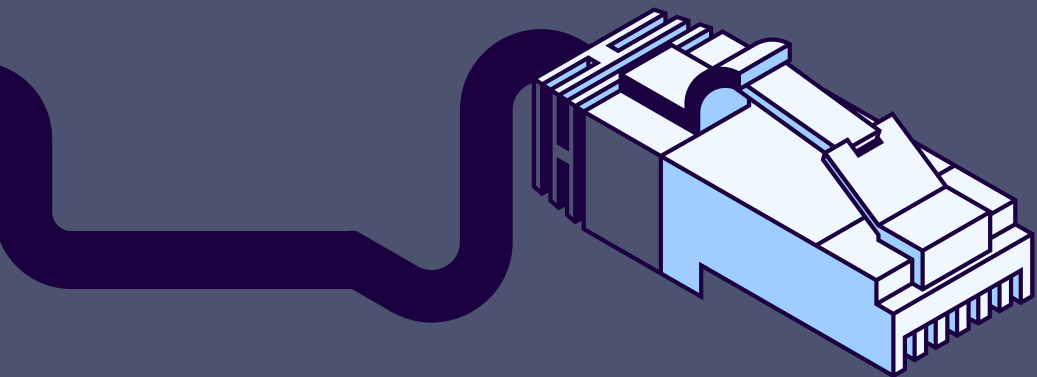
Benefits

Bus networks are cheap and easy to both install and maintain. The use of a single cable means limited hardware is required and new devices can be added easily.

Drawbacks

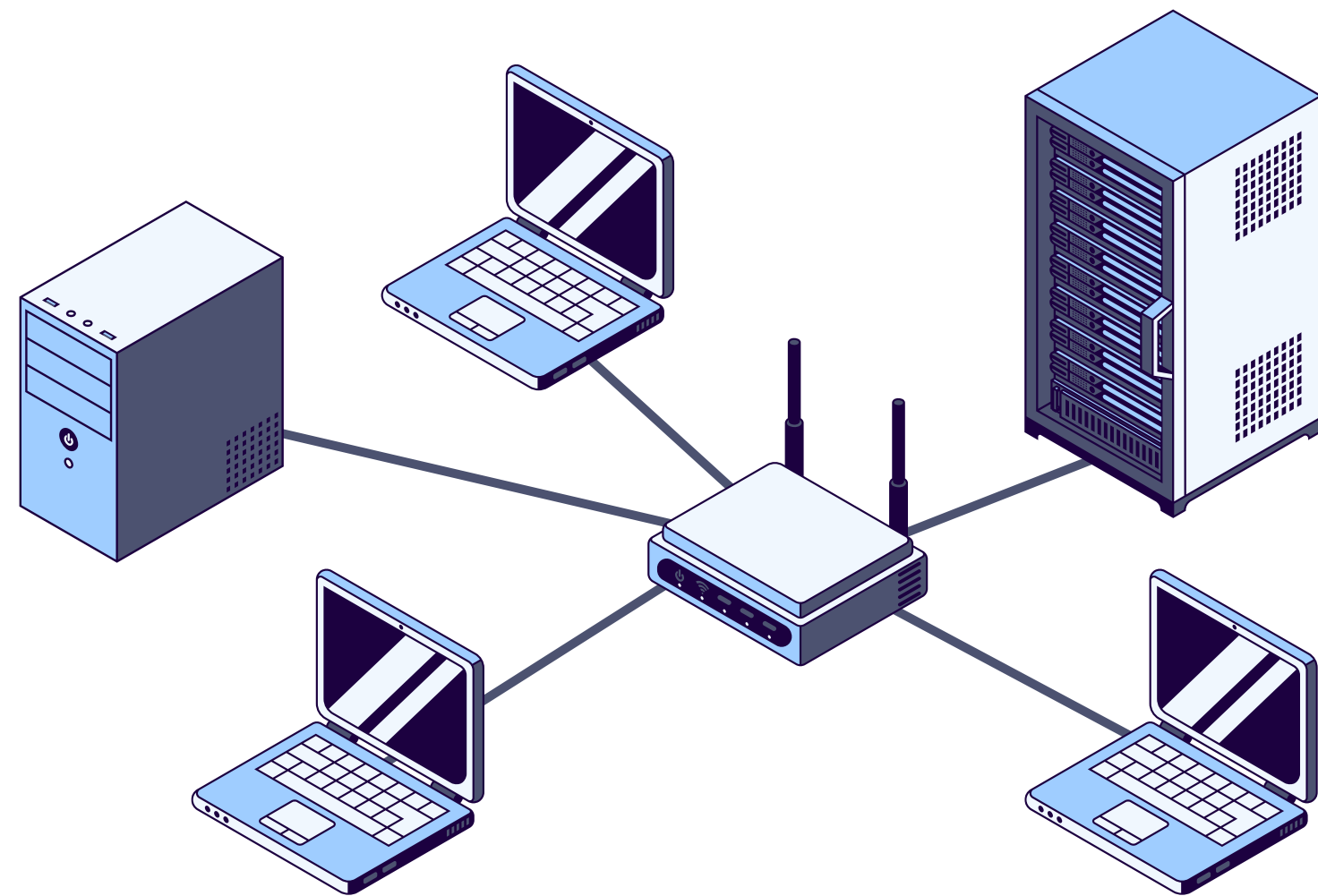
Because it relies on a single cable, if that cable fails, the whole network will fail. This is particularly problematic for larger networks with lots of devices connected.

Bus networks are not typically used for modern networks because of the risk of data collision. When all data is transmitted using one cable, there is a risk that the data will not be delivered correctly.



STAR TOPOLOGY

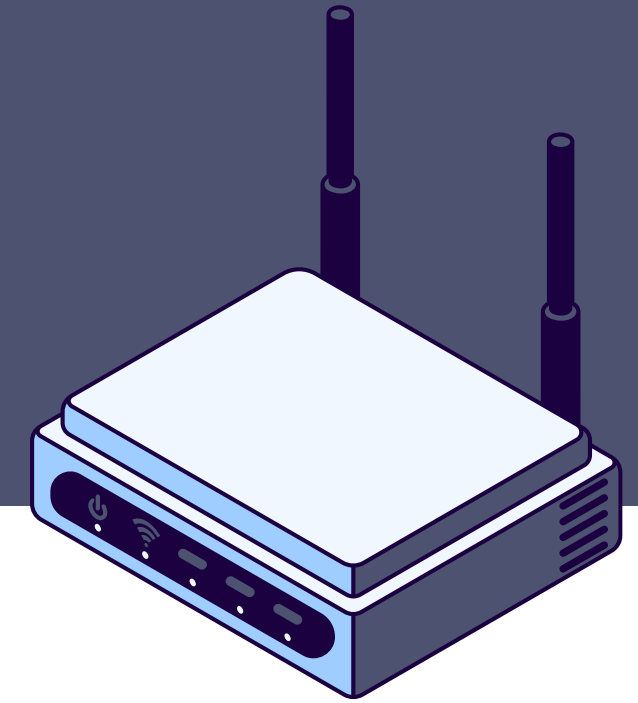
In a star topology, network devices are connected indirectly through a centralised device called a switch. This can be a dedicated device or built into a router on home networks.



Benefits

The star network is more reliable because each device has its own connection to the switch, so a failure of one device does not affect other devices. New devices only need connecting to the switch.

Because star networks route data directly to the correct device, they are much more efficient and the risk for data collision is drastically reduced.



Drawbacks

Star networks are typically used for large permanent networks. Such networks need dedicated hardware (e.g. switch or router) and cables to connect each device. Star networks are therefore more expensive.

USES FOR STAR TOPOLOGY

Sort the characteristics into the correct network topology.

Bus Topology

Star Topology

Uses a centralised switch.

Uses expensive equipment.

Uses a backbone cable.

More reliable and efficient.

Cheap and easy to set up.

Not reliable at large scale.

Used for home networks.

Sort the characteristics into the correct network topology.

Bus Topology

Uses a backbone cable.

Cheap and easy to set up.

Not reliable at large scale.

Star Topology

Uses a centralised switch.

Uses expensive equipment.

More reliable and efficient.

Used for home networks.

For each question, remove the incorrect answer.

Which network is cheapest to set up and maintain?

Bus

Star

Which network is connected and managed centrally?

Bus

Star

Which network routes data to the correct device?

Bus

Star

Which network is more efficient because of reduced data collision?

Bus

Star

Which network can new devices be added to the backbone cable?

Bus

Star

For each question, remove the incorrect answer.

Which network is cheapest to set up and maintain?

Bus

Which network is connected and managed centrally?

Star

Which network routes data to the correct device?

Star

Which network is more efficient because of reduced data collision?

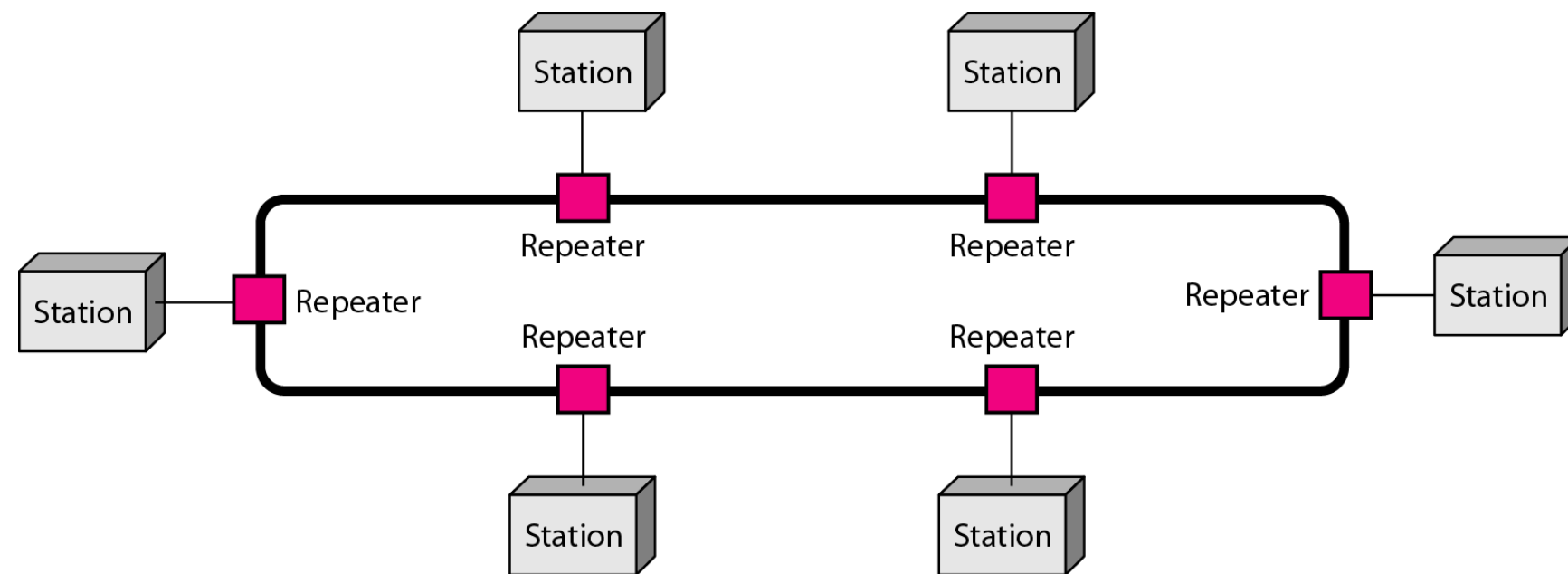
Star

Which network can new devices be added to the backbone cable?

Bus

RING TOPOLOGY

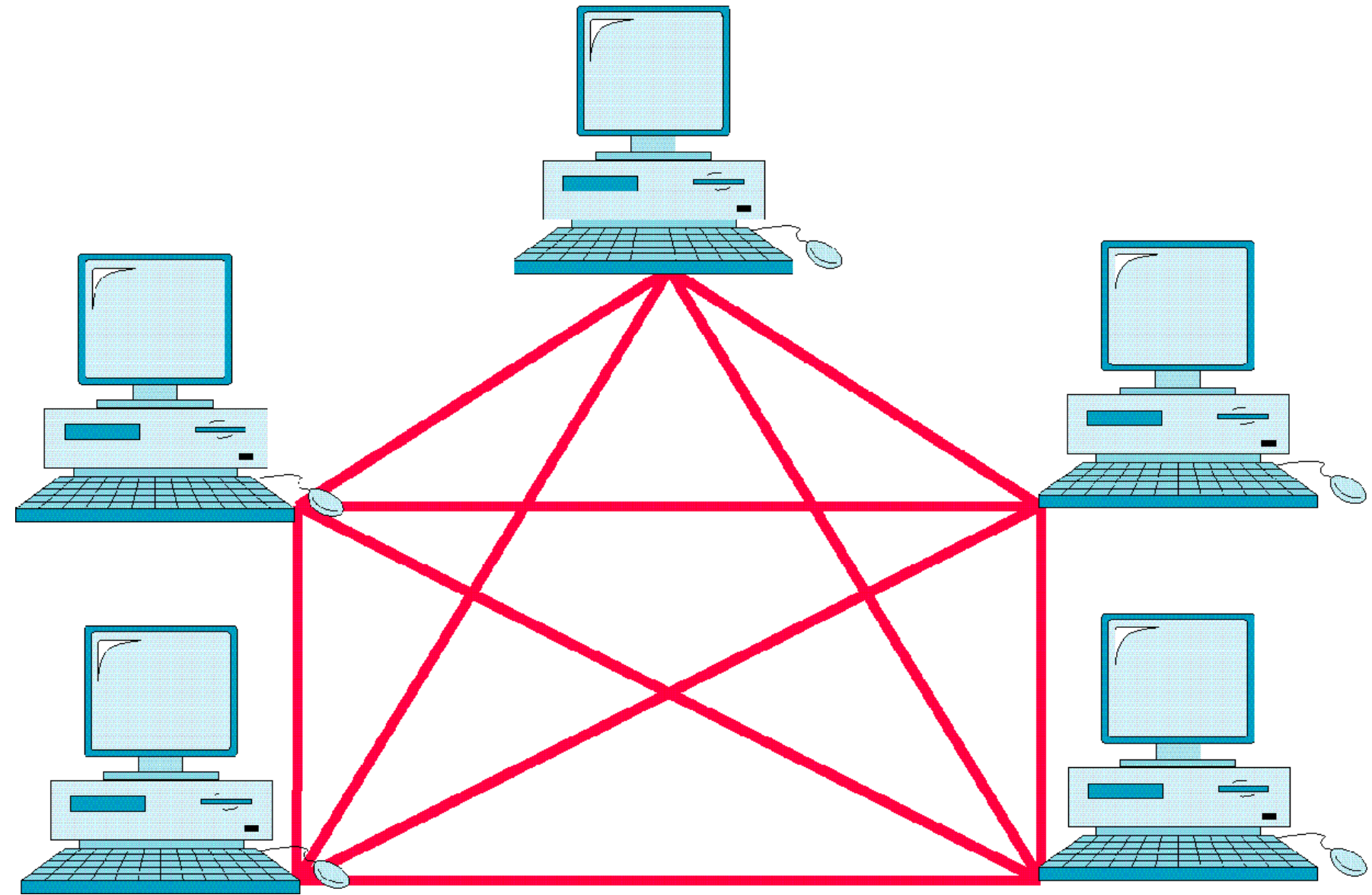
- The ring topology connects workstations in a closed loop.
- Each terminal is connected to TWO other terminals (the next and the previous), with the last terminal being connected to the first.
- Data is transmitted around the ring in one direction only, each station passing on the data to the next station till it reaches its destination.



RING TOPOLOGY

- Ring systems use 4 pair cables (separate send/receive).
- The common implementation of this topology is **token ring**.
- A break in the ring causes the entire network to fail.
 - No central hub.
 - No central cable.
- Network devices connected in a **ring** formation from one device to the next.
- Data travels from one device to another around the ring in one direction only.
- There's no danger of collisions because data always flows in one direction.
- If a connection is broken, the entire network goes down.

MESH TOPOLOGY



MESH TOPOLOGY

- Every computer (node) is directly connected to every other computer in the network.
- Provides multiple redundant communication paths, making the network highly reliable.
- If one link or computer fails, communication continues through alternate paths.
- Commonly used in critical communication systems such as telephone exchanges, military networks, banking networks, and data centers.
- Since every node requires a dedicated connection to every other node, installation cost and cable requirements are very high.

MESH TOPOLOGY

Number of Links Required

In a **full mesh topology**, every node is connected to every other node.

If there are **n** computers:

- Computer 1 connects to **(n – 1)** computers.
- Computer 2 connects to **(n – 2)** new computers.
- Computer 3 connects to **(n – 3)** new computers.

...

Last computer requires **0** new connections.

Total Links

$$(n - 1) + (n - 2) + (n - 3) + \dots + 2 + 1$$

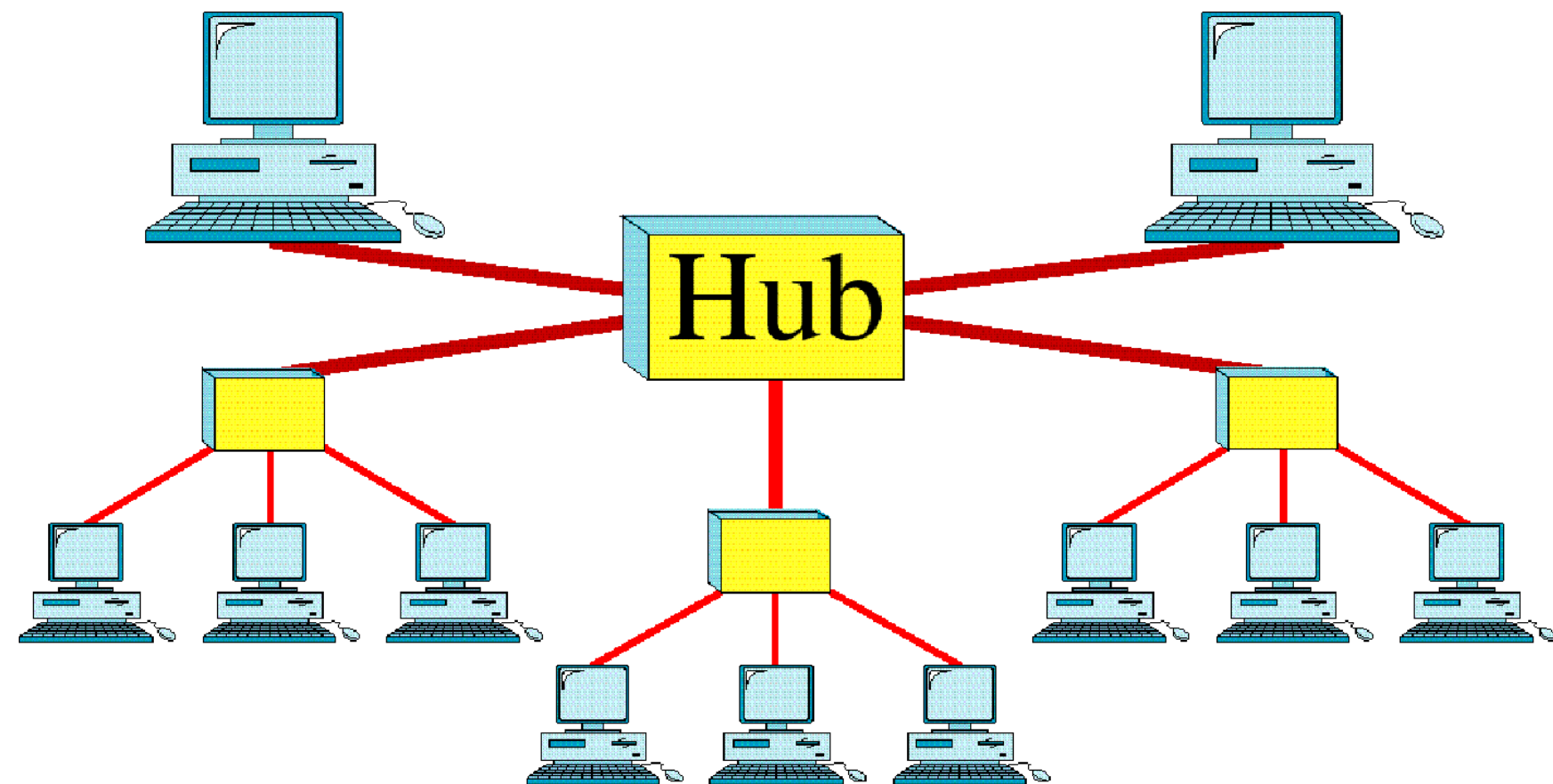
$$\text{Number of Links} = \frac{n(n - 1)}{2}$$

where

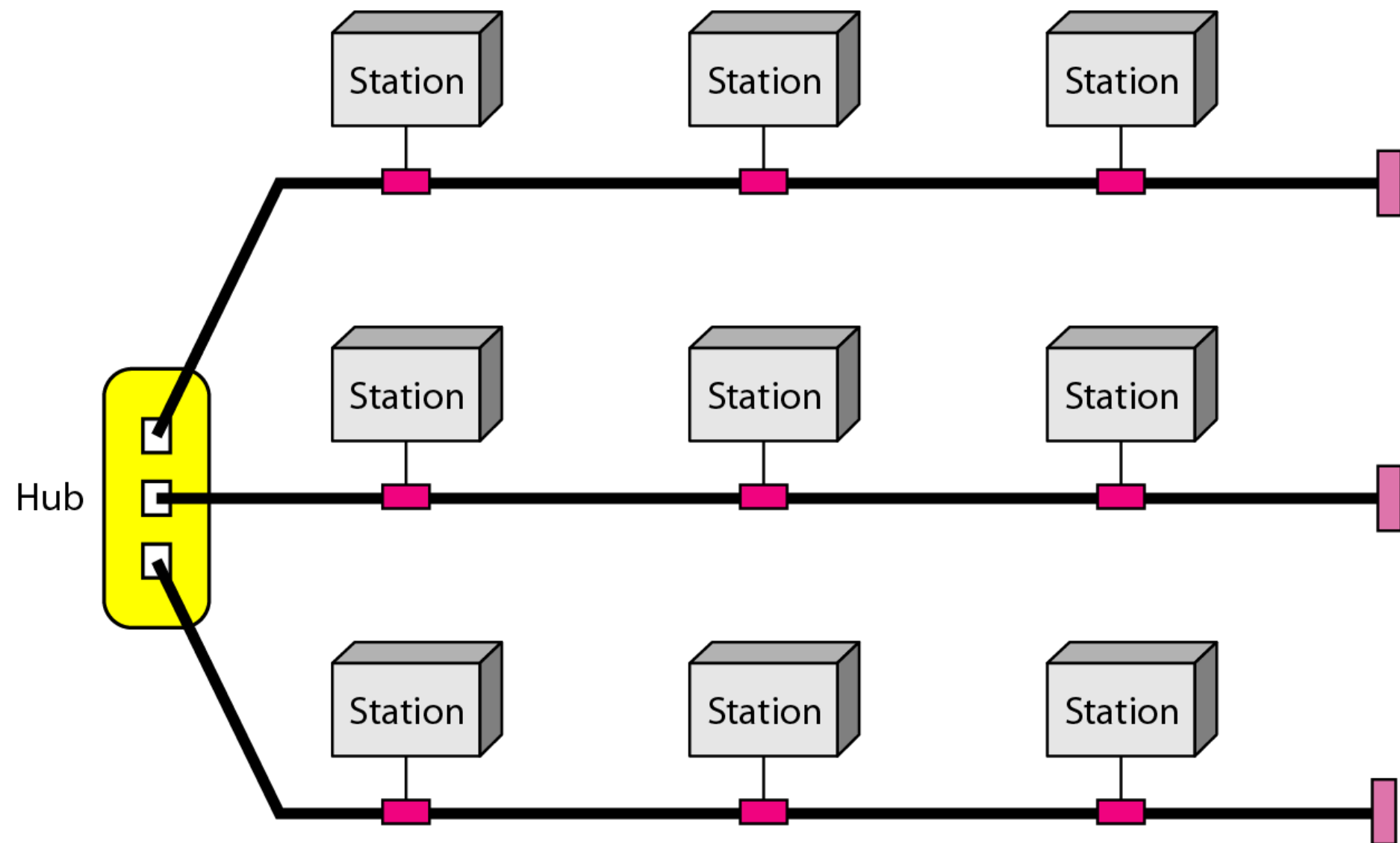
n = Number of computers (nodes)

L = Total number of communication links

TREE TOPOLOGY



HYBRID TOPOLOGY



QUESTION?

Scenario

A new school is setting up a computer laboratory with **40 computers**. The school wants:

- Easy installation and maintenance.
- If one computer fails, the remaining computers should continue working.
- Easy addition of new computers in the future.
- Good performance and simple troubleshooting.

Questions:

- Which network topology would you recommend?
- Justify your choice with at least four reasons.
- Which networking device is required at the center of this topology?

SOLUTION

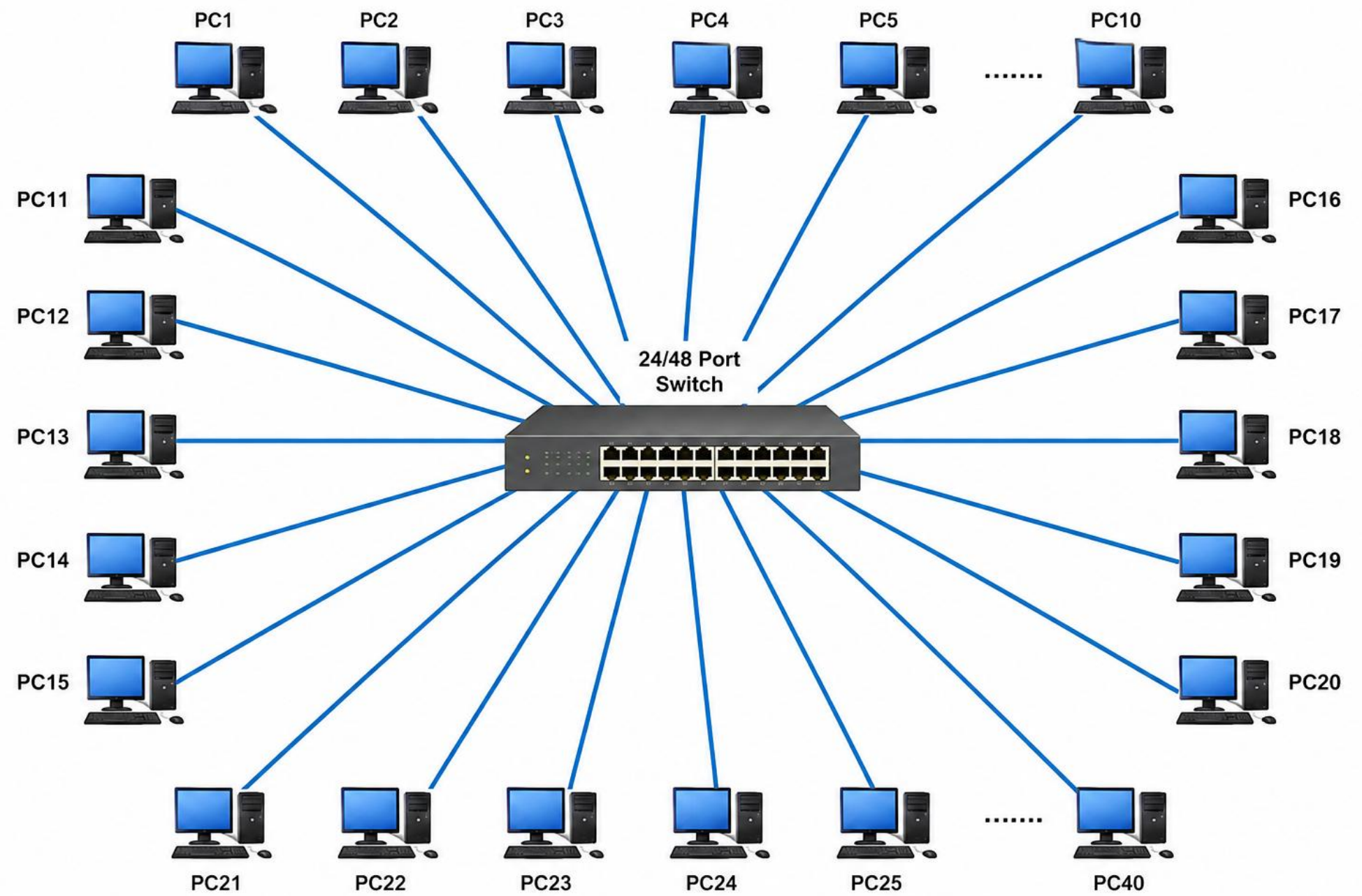
Scenario Analysis

The school is establishing a computer laboratory with 40 computers. The network should satisfy the following requirements:

Requirement	Explanation
Easy installation and maintenance	The network should be simple to set up and manage.
Fault tolerance	Failure of one computer should not affect the others.
Scalability	More computers should be added easily in the future.
Good performance	Data communication should be fast and efficient.
Easy troubleshooting	Faults should be easy to locate and repair.

SOLUTION

Scenario Analysis



SOLUTION

Q1. Which network topology would you recommend?

Recommended Topology: Star Topology

Definition

In a Star Topology, every computer is connected individually to a central networking device (Switch) using separate network cables. All communication between computers passes through the central switch.

SOLUTION

Q2. Justify your choice with at least four reasons.

1. Easy Installation and Maintenance

- Each computer is connected using an independent cable.
- Installation is simple because each connection is separate.
- Maintenance is easier since technicians can work on one computer without affecting others.
- Cable management is organized.
- Benefit for School: The computer laboratory can be installed quickly and maintained easily.

2. Failure of One Computer Does Not Affect Others

If one computer, network cable, or Network Interface Card (NIC) fails: Only that particular computer becomes disconnected. The remaining 39 computers continue working normally.

SOLUTION

Q2. Justify your choice with at least four reasons.

3. Easy Expansion (Scalability)

Suppose the school wants to increase the laboratory from 40 computers to 50 computers.

The administrator simply:

- Connects additional computers to unused switch ports.
- If all ports are occupied, another switch can be connected (uplinked) to expand the network.
- No existing connections need to be modified.

4. Good Performance

A switch provides a dedicated communication path to every connected computer.

Advantages:

- High-speed communication
- Reduced collisions
- Better bandwidth utilization
- Simultaneous communication between multiple computers

SOLUTION

Q2. Justify your choice with at least four reasons.

5. Easy Troubleshooting

Since every computer has an individual cable:

- Faulty cables can be identified easily.
- Switch port LEDs indicate the status of each connection.
- Problems can be isolated quickly.

6. Better Network Management

Modern switches support:

- Port monitoring
- VLANs
- Security features
- Traffic management
- Error detection

This makes administration easier for school network administrators.

SOLUTION

Q3. Which networking device is required at the center of this topology?

Network Switch

Definition: A network switch is a networking device that connects multiple computers in a Local Area Network (LAN) and forwards data only to the intended destination device based on its MAC (Media Access Control) address.

Hub	Switch
Broadcasts data to all computers	Sends data only to the destination computer
More collisions	Almost no collisions
Slower	Faster
Shared bandwidth	Dedicated bandwidth per port
Less secure	More secure

Switch is the preferred central device for modern computer laboratories.

SOLUTION

Q3. Which networking device is required at the center of this topology?

Working of a Switch

1. Computer A sends data.
2. The switch reads the destination MAC address.
3. It looks up its MAC address table.
4. The data is forwarded only to the destination computer.

This improves: Speed, Security, & Network efficiency

Advantages of Using a Switch

- High-speed communication (typically 1 Gbps or higher per port)
- Dedicated bandwidth for each computer
- Reduced network congestion
- Improved security
- Easy expansion by adding more switches
- Supports full-duplex communication (simultaneous sending and receiving)

Parameter	Bus Topology	Star Topology	Ring Topology	Mesh Topology	Tree Topology	Hybrid Topology
Structure	Single backbone cable	All devices connected to a central switch/hub	Devices connected in a closed loop	Every device connected to every other device	Hierarchical arrangement of multiple stars	Combination of two or more topologies
Cable Requirement	Very Low	Moderate	Moderate	Very High	High	Depends on design
Installation Cost	Low	Moderate	Moderate	Very High	High	High
Scalability	Poor	Excellent	Moderate	Poor for large networks	Excellent	Excellent
Performance	Decreases as nodes increase	High due to dedicated links	Predictable but slower than Star	Very High	High	Very High
Fault Tolerance	Very Low	Good	Moderate	Excellent	Good	Excellent
Single Point of Failure	Backbone cable	Central Switch	Ring break (unless dual ring)	No	Root node	Depends on topology
Maintenance	Difficult	Easy	Moderate	Difficult	Easy	Moderate
Troubleshooting	Difficult	Very Easy	Moderate	Difficult	Easy	Moderate
Security	Low	High	Medium	Very High	High	Very High
Network Expansion	Difficult	Easy	Moderate	Very Difficult	Easy	Easy
Reliability	Low	High	Medium	Very High	High	Very High

APPLICATIONS

Topology	Real-Time Example	Industry/Application	Reason for Use
Bus	CAN Bus inside vehicles	Automotive Industry	Low cost, short-distance communication among ECUs
Star	Office LAN, College Computer Lab	IT Companies, Banks, Universities	Easy management, simple troubleshooting, high performance
Ring	SONET/SDH Fiber Ring	Telecom Companies (BSNL, Airtel, Jio Backbone)	Automatic redundancy using dual ring
Mesh	Internet Backbone	ISPs, Google, Amazon AWS, Microsoft Azure	Multiple redundant paths ensure 24×7 availability
Tree	Enterprise Campus Network	Universities, Corporate Offices	Easy departmental expansion and centralized management
Hybrid	Modern Data Centers	Google, Meta, Microsoft, Amazon	Combines performance, scalability, and reliability

EXAMPLES

Sector	Topology Used	Example
Home Network	Star	Wi-Fi Router connected to laptops, mobiles, smart TV
College Network	Tree + Star	Department → Floor Switch → Lab Switch → PCs
Bank	Star	Branch computers connected to central switch
Google Data Center	Hybrid + Mesh	Multiple redundant switches and routers
Internet	Partial Mesh	ISPs interconnected worldwide
Telephone Network	Mesh	Telephone exchanges
Railway Reservation System	Tree	Central server with regional and local servers
Airport Network	Hybrid	Security, CCTV, Wi-Fi, Ticketing
Hospital Network	Star	Nurse stations connected to central server
Factory Automation	Bus	PLC and sensors using CAN Bus/Profibus/Modbus